

STAT214 Chapter 13 Producing the Final Prediction Results

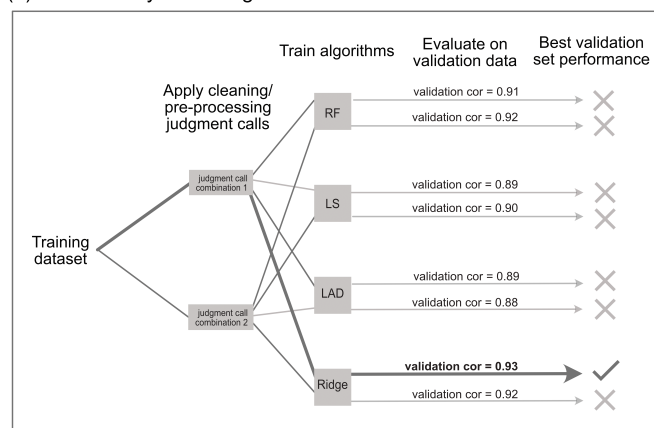
1 Approach 1: Choosing a Single Predictive Fit Using PCS

1.1 Choosing a Single Predictive Fit Using PCS

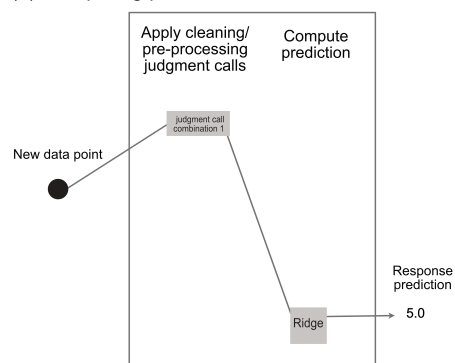
- this PCS-based approach selects the best algorithm across *many* possible versions of the cleaned/preprocessed training datasets
- the PCS approach to choosing the best fit identifies the best *combination* of algorithm and cleaning/preprocessing judgment calls

1.2 The Single Best Fit Approach

(a) Predictability screening



(b) Computing predictions for new observations



1.3 Steps for Choosing a Single PCS Predictive Fit

1. **Predictability screening:** Identify which fit has the best validation set performance.
 1. Create and document several different versions of the cleaned and preprocessed training and validation sets using different combinations of cleaning and preprocessing judgment calls. Let K denote the number of cleaned/preprocessed datasets you end up with.
 2. Train each predictive algorithm (e.g., LS, RF, and so on) using each of the K cleaned/preprocessed training datasets. If you have L different algorithms, then you will end up with $K \times L$ predictive fits.
 3. Generate a response prediction for each *validation set* observation using each of the $K \times L$ predictive fits.
 4. Identify which predictive fit from among the $K \times L$ fits yields the best validation predictive performance. This is the best PCS fit.
2. **Computing predictions for new observations:** Predictions for new observations can be computed using the best PCS fit.
3. **Test set evaluation:** Evaluate your best fit using the *test set* observations to provide a final independent assessment of its

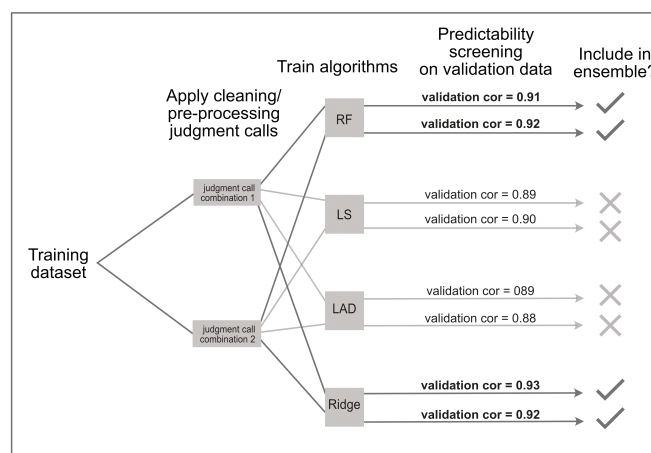
2 Approach 2: PCS Ensemble

2.1 PCS Ensemble

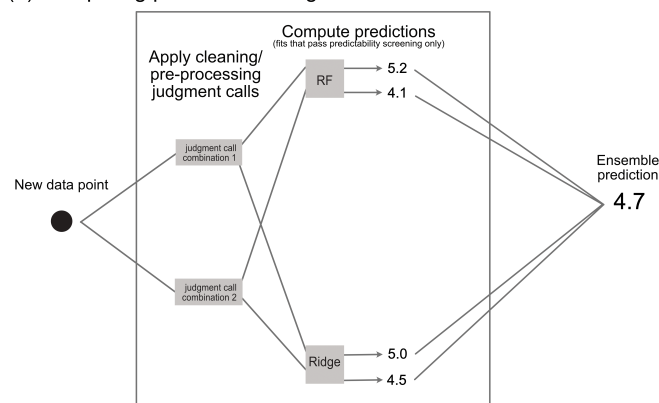
- combine multiple accurate algorithms into an **ensemble**
- By creating an ensemble of several different PCS predictive fits, we can combine the different patterns that are being captured across the individual fits.
- A PCS ensemble generates a prediction for a new (future) data point by generating a separate prediction for each of the individual predictive fits that are to be included in the ensemble (i.e., that pass a relevant predictability screening) and combining their predictions into a single value.

2.2 The Ensemble Approach

(a) Predictability screening



(b) Computing predictions using the ensemble



2.3 Process of Creating an Ensemble

1. **Predictability screening**: Identify which fits will be included in the ensemble.
 1. Create and document several different versions of the cleaned and preprocessed training, validation, and test sets using a range of combinations of cleaning and preprocessing judgment calls. Let K denote the number of cleaned/preprocessed datasets you end up with.
 2. Train each relevant predictive algorithm (e.g., LS, RF, etc.) using each of the K cleaned/preprocessed versions of the training data. If you have L different algorithms, then you will end up with $K \times L$ different predictive fits.
 3. Using each of the $K \times L$ predictive fits, generate a response prediction for each **validation set** observation.
 4. Evaluate the validation set predictive performance for each of the $K \times L$ fits (e.g., using rMSE or correlation for continuous responses and accuracy or AUC for binary responses).
 5. Conduct **predictability screening** by keeping only the best fits (say, the top 10 percent) in terms of **validation set** predictive performance. This threshold (e.g., the top 10 percent of fits) can be based on domain knowledge or it

can be tuned using the ensemble's predictive performance on the validation set.

2. **Compute predictions for new observations using the ensemble:** Predictions from the ensemble can be computed based on the fits that pass the predictability screening step (e.g., the top 10 percent of fits) by averaging their continuous predicted responses or taking the majority class of their binary predicted responses.
3. **Test set evaluation:** Evaluate your ensemble's predictive performance using the **test set** observations to provide a final independent assessment of its predictive performance.

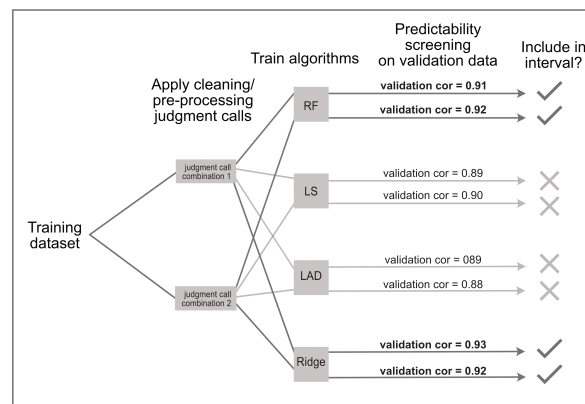
3 Approach 3: Calibrated PCS Prediction Perturbation Intervals

3.1 Prediction Perturbation Intervals (PPIs)

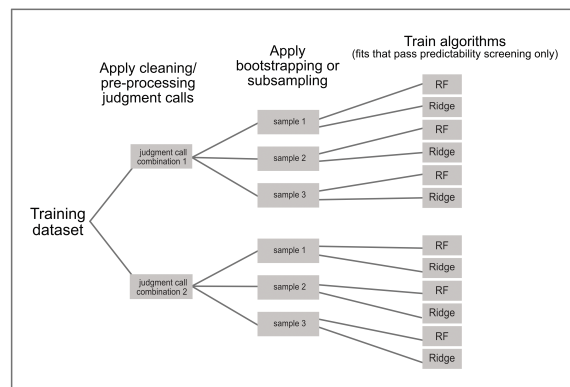
- PPIs correspond to an interval of plausible predictions from among the fits that pass a validation set predictability screening
- The predictability screening procedure for creating PPIs is the same as for the ensemble approach

3.2 The PPI Approach

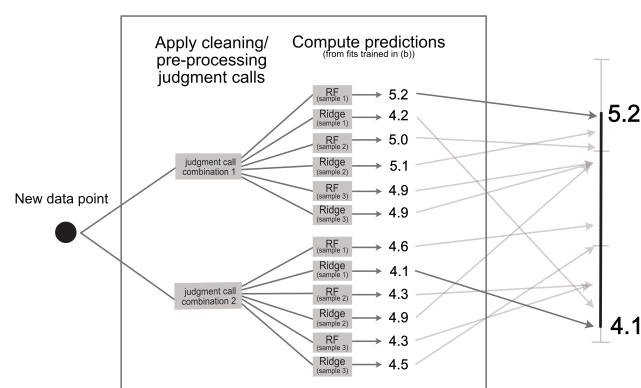
(a) Predictability screening



(b) Bootstrapping and re-training



(c) Computing (uncalibrated) prediction perturbation intervals



3.3 Coverage and Calibration

- The coverage of a collection of PPIs corresponds to the proportion of the intervals that contain the observed response.
- One way to calibrate the PPIs to achieve a particular coverage is to choose a multiplicative factor, γ , by which to expand or shorten the lengths of the intervals such that the modified intervals have a coverage of 90 percent. One possible formula for a set of calibrated PPIs is given by

$$[\text{median} - \gamma \times (\text{median} - q_{0.05}), \text{median} + \gamma \times (q_{0.05} - \text{median})]$$

The value of γ is chosen by trying several possible values of γ and identifying which value yields a calibrated validation set PPI coverage closest to 90 percent.

3.4 Computing Prediction Perturbation Intervals

1. **Predictability screening:** Identify which fits will be included in the interval.
 1. Create and document several different versions of the cleaned and preprocessed training, validation, and test sets using a range of combinations of cleaning and preprocessing judgment calls. Let K denote the number of cleaned/preprocessed datasets you end up with.
 2. Train each relevant predictive algorithm (e.g., LS, RF, etc.) using each of the K cleaned/preprocessed versions of the training data. If you have L different algorithms, then you will end up with $K \times L$ different predictive fits.
 3. Using each of the $K \times L$ predictive fits, generate a response prediction for each *validation set* observation.
 4. Evaluate the validation set predictive performance for each of the $K \times L$ fits (e.g., using rMSE or correlation for continuous responses).
 5. Conduct *predictability screening* by keeping only the best fits (say, the top 10 percent) in terms of *validation set* predictive performance.
2. **Conduct bootstrap perturbations and retrain each relevant fit:**
 1. Create J bootstrapped, data-perturbed versions of each of the cleaned/preprocessed datasets that were involved in fits that passed the predictability screening.
 2. Retrain each of the fits that passed the predictability screening using each of the J bootstrapped versions of the relevant cleaned/preprocessed training datasets. If $x \times K \times L$ fits passed the predictability screening (where x corresponds to the proportion of fits that passed the predictability screening; $0 < x \leq 1$ and $x = 0.1$ corresponds to selecting the top 10 percent of fits), then you will end up with $x \times J \times K \times L$ predictive fits (if this value is not an integer, you can round up to the nearest integer). We recommend ensuring that $x \times J \times K \times L \geq 100$.
3. **Calibrate the PPIs using the validation set:**
 1. If you are aiming for $(1 - \alpha)$ percent coverage, first *compute an uncalibrated interval* for each validation set observation based on the $\alpha/2$ and $(1 - \alpha/2)$ quantiles of the predictions: $[q_{\alpha/2}, q_{(1-\alpha/2)}]$.
 2. Estimate the *coverage* of your uncalibrated validation set intervals by computing the proportion of the intervals that contain the observed response.
 3. Identify the *calibration constant*, γ , the magnitude by which you need to lengthen or shorten the intervals to achieve a validation set coverage of $(1 - \alpha)$ percent. This can be done by computing the coverage of the calibrated intervals using

$$[\text{median} - \gamma \times (\text{median} - q_{\alpha/2}), \text{median} + \gamma \times (q_{1-\alpha/2} - \text{median})]$$

for various values of γ .

4. **Compute calibrated PPIs for new data points:** Computing calibrated PPIs for new data points involves computing the response prediction using each of the $x \times J \times K \times L$ bootstrapped fits that pass the predictability screening, identifying the $\alpha/2$ and $(1 - \alpha/2)$ quantiles ($q_{\alpha/2}$ and $q_{(1-\alpha/2)}$) of these predictions, and plugging these quantiles and the value of γ identified in the previous calibration step into the calibrated PPI equation:

$$[\text{median} - \gamma \times (\text{median} - q_{\alpha/2}), \text{median} + \gamma \times (q_{1-\alpha/2} - \text{median})]$$

5. **Test set evaluation:** Compute calibrated PPIs for each test set observation and compute their coverage.